

Name :

Roll No. :

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Assignment 1

Vth Semester

AIT307 - INTRODUCTION TO ARTIFICIAL INTELLIGENCE

AI & DS 2023

Total Mark: 15

Total Time: 5 days

Course Outcome (CO)	
After completion of this course, the students will be able to	
CO2	Apply uninformed and informed search techniques for problem solving in intelligent systems.
CO3	Solve Constraint Satisfaction Problems using search techniques.

No.	Question	Marks	CO	BL
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1. "Gymnasiums are full of different exercise equipment with different calorie burn rates. When doing exercise, it is important to cover the full-body, rather than a part of the body. Selecting exercise techniques with the least recovery time can reduce the overall time taken to do the exercise as well as any damages to the body. Therefore selecting the right exercise machine to burn the target calorie, and involving the whole body in to exercise process and with least prone to cause injuries is important. The scenario is based on the information given in table 1 and state-space."

Different exercise equipment vs calorie burn is given in the table.

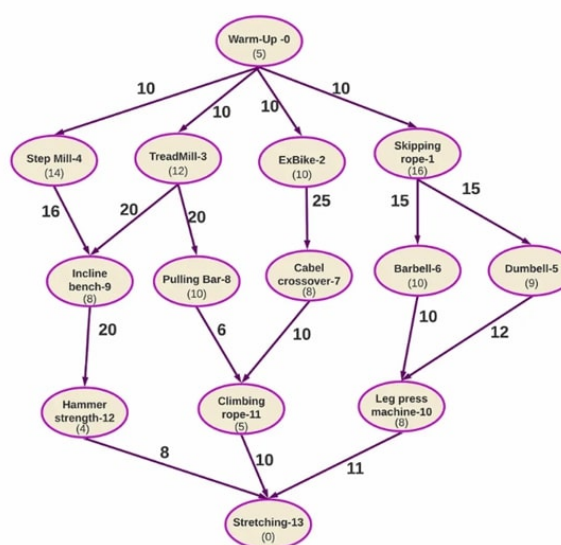
Path Cost: Time taken to burn 300 Cal (minutes)

Heuristic Value: Recovery time required after burning 300 Cal (minutes)

The given figure shows the state space for the problem.

- a) Apply uninformed (BFS, DFS, DLS ,UCS, IDS) and informed(Greedy BFS and A*) search algorithms to the given state-space problem and show the traversal through the state space.
- b) Compare the features including Completeness, Optimality, Space and Time Complexity.

Exercise activity / machine	Coverage of different parts of the body	Time taken to burn 300 Cal (Minutes)	Recovery time after burning 300 Cal (Minutes)
0. Warmup activities	Full body	10	5
1. Skipping rope	Upper + Lower body	15	16
2. Exercise bike		25	10
3. Tread Mill	Lower body	20	12
4. Step Mill		16	14
5. Dumbbell		12	9
6. Barbell	Upper body	10	10
7. Cable-Crossover		10	8
8. Pulling bars	Upper + Middle body	6	10
9. Incline bench		20	8
10. Leg press machine	Middle body (abdomen)	11	8
11. Climbing rope	Upper + Middle + Lower body	10	5
12. Hammer strength	Upper body	8	4
13. Stretching	Full body	-	0



10 CO2 3

2.	Consider the following game tree. Cross out the branches that do not need to be examined by alpha-beta search in order to find the minimax value of the top node.			
		5	CO3	3

Question Paper Level	3
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Prepared by	Verified by	Approved by

LEVEL - Bloom's Taxonomy Level
Level 3 Applying(P)